

- The last 2 doses of paliperidone Sustenna should ideally be the same dosage strength, so that a consistent maintenance dose is established prior to starting paliperidone Trinza
- Conversion from 1-month injectable: initiate 3-month LAI when the next 1-month LAI injection is scheduled; dosing is based on the previous 1-month injection dose

1-month Sustenna	3-month Trinza
78 mg	273 mg
117 mg	410 mg
156 mg	546 mg
234 mg	819 mg

- Dose adjustments can be made every 3 months if needed; the long-acting nature of paliperidone Trinza means that a patient's response to an adjusted dose may not be apparent for multiple months
- Injection can be given up to 2 weeks before or after the 3-month time point
- If dose has been missed for 4–9 months, the following reinitiation schedule must be followed:

Last Trinza Dose	Day 1	Day 8	1 Month After Day 8
-------------------------	--------------	--------------	----------------------------

Last Trinza Dose	Day 1	Day 8	1 Month After Day 8
273 mg	78 mg Sustenna (deltoid)	78 mg Sustenna (deltoid)	273 mg Trinza (deltoid or gluteal)
410 mg	117 mg Sustenna (deltoid)	117 mg Sustenna (deltoid)	410 mg Trinza (deltoid or gluteal)
546 mg	156 mg Sustenna (deltoid)	156 mg Sustenna (deltoid)	546 mg Trinza (deltoid or gluteal)
819 mg	156 mg Sustenna (deltoid)	156 mg Sustenna (deltoid)	819 mg Trinza (deltoid or gluteal)

- If dose has been missed for more than 9 months, reinitiate treatment with paliperidone Sustenna according to its prescribing information; paliperidone Trinza can be used after the patient has been adequately treated with paliperidone Sustenna for at least 4 months

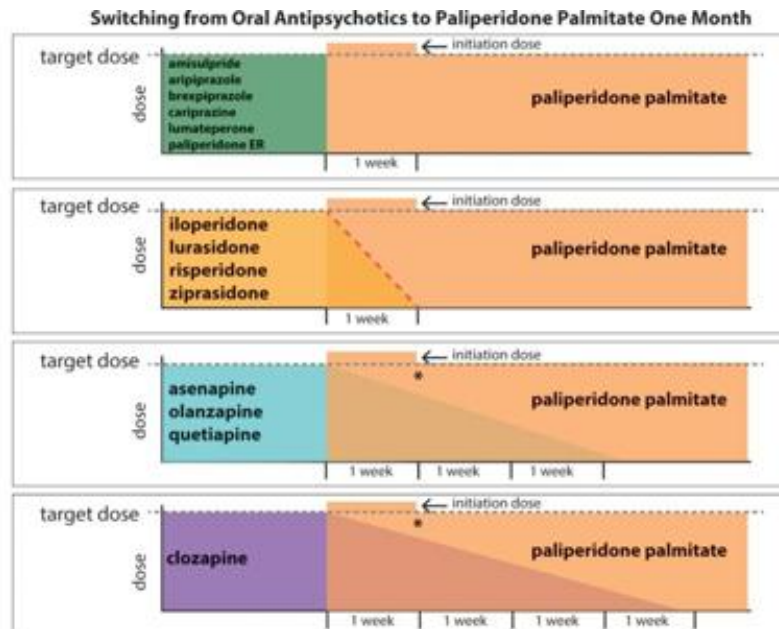
Dosing Tips

- With LAIs, the absorption rate constant is slower than the elimination rate constant, thus resulting in “flip-flop” kinetics –i.e., time to steady state is a function of absorption rate, while concentration at steady state is a function of elimination rate

- The rate-limiting step for plasma drug levels for LAIs is not drug metabolism, but rather slow absorption from the injection site
- In general, 5 half-lives of any medication are needed to achieve 97% of steady-state levels
- The long half-lives of depot antipsychotics mean that one must either adequately load the dose (if possible) or provide oral supplementation
- The failure to adequately load the dose leads either to prolonged cross-titration from oral antipsychotic or to sub-therapeutic antipsychotic plasma levels for weeks or months in patients who are not receiving (or adhering to) oral supplementation
- Because plasma antipsychotic levels increase gradually over time, dose requirements may ultimately decrease from initial; obtaining periodic plasma levels can be beneficial to prevent unnecessary plasma level creep
- The time to get a blood level for patients receiving LAI is the morning of the day they will receive their next injection
- Kinetics for paliperidone palmitate are determined by particle size: smaller particles (1-month) vs. larger particles (3-month)
- Advantages: no need for oral coverage; 3-month injection schedule with Trinza
- Disadvantages: plasma levels have limited value in guiding treatment

The Art of Switching

🔄 Switching from Oral Antipsychotics to Paliperidone Palmitate (1-Month)



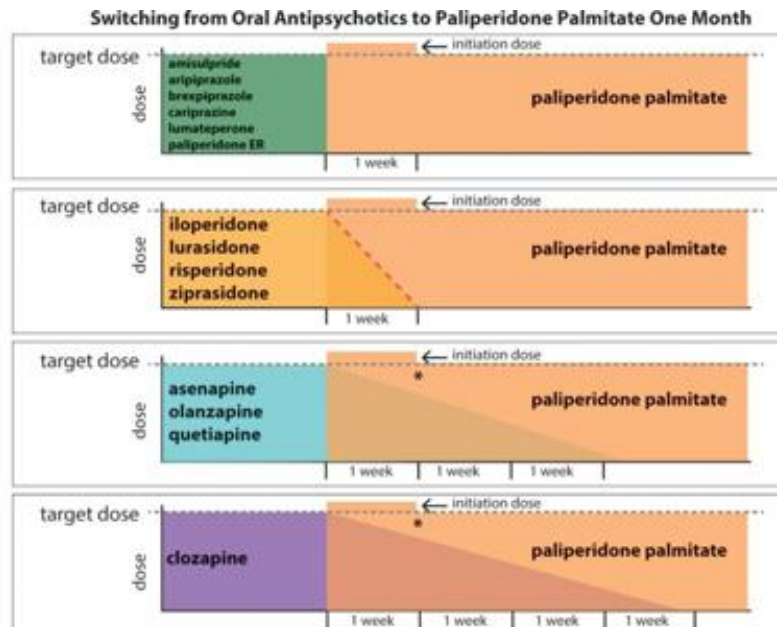
- Discontinuation of oral antipsychotic can begin immediately if adequate loading is pursued
- How to discontinue oral formulations
 - Down-titration is not required for: amisulpride, aripiprazole, brexpiprazole, cariprazine, lumateperone, or paliperidone ER
 - 1-week down-titration is required for: iloperidone, lurasidone, risperidone, ziprasidone
 - 3–4-week down-titration is required for: asenapine, olanzapine, quetiapine
 - 4+-week down-titration is required for: clozapine

- For patients taking benzodiazepine or anticholinergic medication, this can be continued during cross-titration to help alleviate side effects such as insomnia, agitation, and/or psychosis. Once the patient is stable on LAI, these can be tapered one at a time as appropriate.

The Art of Switching ↔ Switching from Oral Antipsychotics to Paliperidone ER

- Due to OROS technology, paliperidone ER can be initiated at full desired dose; however, titration over 1–2 weeks may be appropriate for some patients
 - With amisulpride, aripiprazole, brexpiprazole, cariprazine, and lumateperone, immediate stop is possible; begin paliperidone ER at an intermediate, or if needed, effective dose
 - Risperidone, ziprasidone, iloperidone, and lurasidone can be tapered off over a period of 1 week due to the risk of withdrawal symptoms such as insomnia
 - Clinical experience has shown that quetiapine, olanzapine, and asenapine should be tapered off slowly over a period of 3–4 weeks, to allow patients to readapt to the withdrawal of blocking cholinergic, histaminergic, and alpha 1 receptors
 - Clozapine should always be tapered off slowly, over a period of 4 weeks or more
- * Benzodiazepine or anticholinergic medication can be administered during cross-titration to help alleviate side effects such as insomnia,

agitation, and/or psychosis



Suggested Reading

Harrington CA , English C . Tolerability of paliperidone: a meta-analysis of randomized, controlled trials . *Int Clin Psychopharmacol* 2010 ;**25** (6):334–41 .

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

Nussbaum A , Stroup TS . Paliperidone for schizophrenia . *Cochrane Database Syst Rev* 2008 ;**16** (2):CD006369 .

.....
Nussbaum AM , Stroup TS . Paliperidone palmitate for schizophrenia .
Cochrane Database Syst Rev . 2012 ;(6):CD008296 .
.....

Patel C , Emond B , Lafeuille MH , et al. Real-world analysis of switching
patients with schizophrenia from oral risperidone or oral paliperidone to
once-monthly paliperidone palmitate . *Drugs Real World Outcomes* 2020 ;7
(1):19 –29 . doi: 10.1007/s40801-019-00172-9 .

Paroxetine

Therapeutics

Brands

- Paxil
- Paxil CR
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin reuptake inhibitor (S-RI)
- SSRI (selective serotonin reuptake inhibitor); often classified as an antidepressant, but it is not just an antidepressant

Commonly Prescribed for

(bold for FDA approved)

- **Major depressive disorder (paroxetine and paroxetine CR)**
- **Obsessive-compulsive disorder (OCD)**
- **Panic disorder (paroxetine and paroxetine CR)**

- **Social anxiety disorder (social phobia) (paroxetine and paroxetine CR)**
- **Posttraumatic stress disorder (PTSD)**
- **Generalized anxiety disorder (GAD)**
- **Premenstrual dysphoric disorder (PMDD) (paroxetine CR)**
- **Vasomotor symptoms (Brisdelle)**

How the Drug Works

- Boosts neurotransmitter serotonin
- Blocks serotonin reuptake pump (serotonin transporter)
- Desensitizes serotonin receptors, especially serotonin 1A autoreceptors
- Presumably increases serotonergic neurotransmission
- Paroxetine also has mild anticholinergic actions
- Paroxetine may have mild norepinephrine reuptake blocking actions

How Long Until It Works

- * Some patients may experience relief of insomnia or anxiety early after initiation of treatment
- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks

- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- By contrast, for generalized anxiety, onset of response and increases in remission rates may still occur after 8 weeks of treatment and for up to 6 months after initiating dosing
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced (e.g., OCD, PTSD)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and

problems concentrating in depression)

- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- Trazodone, especially for insomnia
- Bupropion, mirtazapine, reboxetine, or atomoxetine (add with caution and at lower doses since paroxetine could theoretically raise atomoxetine levels); use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation
- Modafinil, especially for fatigue, sleepiness, and lack of concentration

- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, treatment-resistant depression, or treatment-resistant anxiety disorders
- Benzodiazepines
- If all else fails for anxiety disorders, consider gabapentin or tiagabine
- Hypnotics for insomnia
- Classically, lithium, buspirone, or thyroid hormone

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in serotonin concentrations at serotonin receptors in parts of the brain and body other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of serotonin in the gut causing diarrhea, etc.)
- Increasing serotonin can cause diminished dopamine release and might contribute to emotional flattening, cognitive slowing, and apathy in some patients

- Most side effects are immediate but often go away with time, in contrast to most therapeutic effects, which are delayed and are enhanced over time
- * Paroxetine's weak antimuscarinic properties can cause constipation, dry mouth, sedation

Notable Side Effects

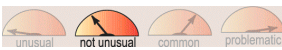
- Sexual dysfunction (dose-dependent; men: delayed ejaculation, erectile dysfunction; men and women: decreased sexual desire, anorgasmia)
- Gastrointestinal (decreased appetite, nausea, diarrhea, constipation, dry mouth)
- Mostly CNS (insomnia but also sedation, agitation, dose-dependent tremors, headache, dizziness)
- Weight gain
- Activation (short-term; patients with diagnosed or undiagnosed bipolar or psychotic disorders may be more vulnerable to CNS-activating actions of SSRIs)
- Autonomic (dose-dependent sweating)
- Bruising and rare bleeding
- SIADH (syndrome of inappropriate antidiuretic hormone secretion)



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Occurs in significant minority

Sedation



- Many experience and/or can be significant in amount
- Generally transient

What to Do About Side Effects

- Wait
- Wait
- Wait
- If paroxetine is sedating, take at night to reduce daytime drowsiness
- Reduce dose to 5–10 mg (12.5 mg for CR) until side effects abate, then increase as tolerated, usually to at least 20 mg (25 mg CR)
- In a few weeks, switch or add other drugs

Best Augmenting Agents for Side Effects

- Often best to try another SSRI or another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for insomnia
- Bupropion, sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Bupropion for emotional flattening, cognitive slowing, or apathy
- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Benzodiazepines for jitteriness and anxiety, especially at initiation of treatment and especially for anxious patients
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of paroxetine

Dosing and Use

Usual Dosage Range

- Depression: 20–50 mg (25–62.5 mg CR)
- Vasomotor symptoms: 7.5 mg at bedtime
- Anxiety disorders and OCD: 10–60 mg/day (12.5–75 mg CR)

Dosage Forms

- Tablet 10 mg scored, 20 mg scored, 30 mg, 40 mg
- Controlled-release tablet 12.5 mg, 25 mg, 37.5 mg
- Capsule 7.5 mg
- Liquid 10 mg/5mL –250 mL bottle

How to Dose

- Depression: initial 20 mg (25 mg CR); usually wait a few weeks to assess drug effects before increasing dose, but can increase by 10 mg/day (12.5 mg/day CR) once a week; maximum generally 50 mg/day (62.5 mg/day CR); single dose
- Panic disorder: initial 10 mg/day (12.5 mg/day CR); usually wait a few weeks to assess drug effects before increasing dose, but can increase by 10 mg/day (12.5 mg/day CR) once a week; maximum generally 60 mg/day (75 mg/day CR); single dose
- Social anxiety disorder: initial 20 mg/day (25 mg/day CR); usually wait a few weeks to assess drug effects before increasing dose, but can increase by 10 mg/day (12.5 mg/day CR) once a week; maximum 60 mg/day (75 mg/day CR); single dose

- Other anxiety disorders: initial 20 mg/day (25 mg/day CR); usually wait a few weeks to assess drug effects before increasing dose, but can increase by 10 mg/day (12.5 mg/day CR) once a week; maximum 60 mg/day (75 mg/day CR); single dose
- Premenstrual dysphoric disorder: initial dose 12.5 mg/day CR; should be administered as a single dose in the morning, with or without food; can be administered either daily or only during the luteal phase of the menstrual cycle



Dosing Tips

- 20 mg tablet is scored, so to save costs, give 10 mg as half of 20 mg tablet, since 10 mg and 20 mg tablets cost about the same in many markets
- Given once daily, often at bedtime, but any time of day tolerated
- 20 mg/day (25 mg/day CR) is often sufficient for patients with social anxiety disorder and depression
- Other anxiety disorders, as well as difficult cases in general, may require higher dosing
- Occasional patients are dosed above 60 mg/day (75 mg/day CR), but this is for experts and requires caution
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder and switch to a mood stabilizer or an atypical antipsychotic

- Liquid formulation easiest for doses below 10 mg when used for cases that are very intolerant to paroxetine or especially for very slow down-titration during discontinuation for patients with withdrawal symptoms
 - Paroxetine CR tablets not scored, so chewing or cutting in half can destroy controlled-release properties
 - Unlike other SSRIs and antidepressants where dosage increments can be double and triple the starting dose, paroxetine's dosing increments are in 50% increments (i.e., 20, 30, 40; or 25, 37.5, 50 CR)
 - Paroxetine inhibits its own metabolism and thus plasma concentrations can double when oral doses increase by 50%; plasma concentrations can increase 2–7-fold when oral doses are doubled
- * Main advantage of CR is reduced side effects, especially nausea and perhaps sedation, sexual dysfunction, and withdrawal
- * For patients with severe problems discontinuing paroxetine, dosing may need to be tapered over many months (i.e., reduce dose by 1% every 3 days by crushing tablet and suspending or dissolving in 100 mL of fruit juice and then disposing of 1 mL and drinking the rest; 3–7 days later, dispose of 2 mL, and so on). This is both a form of very slow biological tapering and a form of behavioral desensitization (not for CR)
- For some patients with severe problems discontinuing paroxetine, it may be useful to add an SSRI with a long half-life, especially

fluoxetine, prior to taper of paroxetine; while maintaining fluoxetine dosing, first slowly taper paroxetine and then taper fluoxetine

- Be sure to differentiate between reemergence of symptoms requiring reinstitution of treatment and withdrawal symptoms

Overdose

- Rarely lethal in monotherapy overdose; vomiting, sedation, heart rhythm disturbances, dilated pupils, dry mouth

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects (dizziness, nausea, stomach cramps, sweating, tingling, dysesthesias)
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation
- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly
- * Withdrawal effects can be more common or more severe with paroxetine than with some other SSRIs

- Paroxetine's withdrawal effects may be related in part to the fact that it inhibits its own metabolism
- Thus, when paroxetine is withdrawn, the rate of its decline can be faster as it stops inhibiting its metabolism
- Controlled-release paroxetine may slow the rate of decline and thus reduce withdrawal reactions in some patients
- Readaptation of cholinergic receptors after prolonged blockade may contribute to withdrawal effects of paroxetine

Pharmacokinetics

- Inactive metabolites
- Half-life approximately 24 hours
- Inhibits CYP450 2D6

Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Can increase TCA levels; use with caution with TCAs or when switching from a TCA to paroxetine
- Can cause a fatal "serotonin syndrome" when combined with MAOIs, so do not use with MAOIs or for at least 14 days after MAOIs are stopped

- Do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing paroxetine
- May displace highly protein-bound drugs (e.g., warfarin)
- There are reports of elevated theophylline levels associated with paroxetine treatment, so it is recommended that theophylline levels be monitored when these drugs are administered together
- May increase anticholinergic effects of procyclidine and other drugs with anticholinergic properties
- Can rarely cause weakness, hyperreflexia, and incoordination when combined with sumatriptan or possibly with other triptans, requiring careful monitoring of patient
- Possible increased risk of bleeding, especially when combined with anticoagulants (e.g., warfarin, NSAIDs)
- NSAIDs may impair effectiveness of SSRIs
- Via CYP450 2D6 inhibition, paroxetine could theoretically interfere with the analgesic actions of codeine, and increase the plasma levels of some beta blockers and of atomoxetine
- Via CYP450 2D6 inhibition, paroxetine could theoretically increase concentrations of thioridazine and cause dangerous cardiac arrhythmias
- Paroxetine increases pimozone levels, and pimozone prolongs QT interval, so concomitant use of pimozone and paroxetine is contraindicated

Other Warnings/Precautions

- Add or initiate other antidepressants with caution for up to 2 weeks after discontinuing paroxetine
- Use with caution in patients with history of seizures
- Use with caution in patients with bipolar disorder unless treated with concomitant mood-stabilizing agent
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI
- If patient is taking thioridazine
- If patient is taking pimozide
- If patient is taking tamoxifen

- If there is a proven allergy to paroxetine

Special Populations

Renal Impairment

- Lower dose [initial 10 mg/day (12.5 mg CR), maximum 40 mg/day (50 mg/day CR)]

Hepatic Impairment

- Lower dose [initial 10 mg/day (12.5 mg CR), maximum 40 mg/day (50 mg/day CR)]

Cardiac Impairment

- Preliminary research suggests that paroxetine is safe in these patients
- Treating depression with SSRIs in patients with acute angina or following myocardial infarction may reduce cardiac events and improve survival as well as mood

Elderly

- Lower dose [initial 10 mg/day (12.5 mg CR), maximum 40 mg/day (50 mg/day CR)]
- Risk of SIADH with SSRIs is higher in the elderly

- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older

Children and Adolescents

- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment with antidepressants and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment
- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients
- Not specifically approved, but preliminary evidence suggests possible efficacy in children and adolescents with OCD or social phobia

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Not generally recommended for use during pregnancy, especially during first trimester
- Epidemiological data have shown an increased risk of cardiovascular malformations (primarily ventricular and atrial septal defects) in infants born to women who took paroxetine during the first trimester (absolute risk is small)
- Unless the benefits of paroxetine to the mother justify continuing treatment, consider discontinuing paroxetine or switching to another antidepressant
- Paroxetine use late in pregnancy may be associated with higher risk of neonatal complications, including respiratory distress
- At delivery there may be more bleeding in the mother and transient irritability or sedation in the newborn
- Must weigh the risk of treatment (first trimester fetal development, third trimester newborn delivery) to the child against the risk of no treatment (recurrence of depression, maternal health, infant bonding) to the mother and child
- For many patients this may mean continuing treatment during pregnancy
- SSRI use beyond the 20th week of pregnancy may be associated with increased risk of pulmonary hypertension in newborns, although this is not proven
- Exposure to SSRIs late in pregnancy may be associated with increased risk of gestational hypertension and preeclampsia